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Business Context:

Over-the-top (OTT) platforms deliver media content directly to users over the internet without requiring traditional cable or satellite services. These platforms provide on-demand access to movies, TV shows, and original content through websites, mobile applications, and smart TVs.

With increasing internet penetration and changing consumer preferences, OTT platforms have experienced rapid growth worldwide. Viewers are increasingly shifting from traditional broadcasting services to on-demand streaming platforms due to their flexibility, convenience, and wide content availability. To remain competitive in this fast-growing market, OTT platforms must understand the factors that influence content engagement and viewer behaviour.

Problem Definition:

ShowTime is an OTT streaming platform that releases various types of content such as movies and web series. The company wants to identify the key factors that influence first-day viewership of content on its platform. Understanding these drivers will help the company optimize its content release strategy, marketing campaigns, and scheduling decisions to improve audience engagement.

The analysis aims to answer the following questions:

- Which variables are the most important predictors of first-day content viewership?
- What does the distribution of first-day content views look like?
- How is content distributed across different genres?
- How does viewership vary by day of release?
- How does seasonality affect content viewership?
- What is the relationship between trailer views and content views?

Data Background:

The dataset contains information about various factors that may influence first-day content viewership on the ShowTime platform. Each observation represents a content release along with several attributes describing its promotion, timing, and audience engagement.

- The dataset contains 1000 rows and 8 columns representing various factors influencing first-day content viewership.
- The columns `visitors`, `ad_impressions`, `views_trailer`, and `views_content` are numerical variables representing platform traffic, marketing reach, trailer popularity, and actual content viewership respectively.
- The column `major_sports_event` is a binary categorical variable (0 or 1) indicating whether a major sports event occurred on the same day as the content release.
- The columns `genre`, `dayofweek`, and `season` are categorical variables that describe the type of content and the timing of the release. These variables could require encoding before building the regression model.

- The target variable for the predictive model is `views_content`, which represents the first-day viewership (in millions).
- There are no missing values in the dataset, which simplifies the preprocessing stage since no imputation is required.
- Since the dataset contains both numerical and categorical variables, we will need to perform EDA separately for numerical distributions and categorical frequencies before building the regression model.

The goal of this analysis is to use these variables to build a linear regression model that identifies the key drivers of first-day content viewership.

Data Preprocessing:

Checking Datatypes:

```
visitors          float64
ad_impressions    float64
major_sports_event  int64
genre             object
dayofweek         object
season            object
views_trailer      float64
views_content      float64
dtype: object
```

Datatypes of dataset

- `visitors`, `ad_impressions`, `views_trailer`, and `views_content` are numerical variables stored as `float64`.
- `major_sports_event` is an integer variable representing a binary indicator (0 or 1).
- `genre`, `dayofweek`, and `season` are categorical variables stored as `object` type.
- The categorical variables will need to be converted to dummy variables before building the regression model.

Unique & Duplicate Values:

```
visitors          56
ad_impressions    80
major_sports_event  2
genre             8
dayofweek         7
season            4
views_trailer      799
views_content      61
dtype: int64
```

Unique values of dataset

- No duplicate rows were found in the dataset.
- Therefore, no rows are needed to be removed.

Checking for Missing Values:

```
visitors          0
ad_impressions    0
major_sports_event 0
genre             0
dayofweek         0
season            0
views_trailer     0
views_content     0
dtype: int64
```

Missing Value Count from Dataset

- There are no missing values in any of the columns of the ShowTime dataset.
- All 8 variables have complete data, so no missing value treatment or imputation is required.

Inspecting categorical variables:

- The variable major_sports_event showed that 600 releases occurred without a major sports event, while 400 releases coincided with a sports event.
- The genre distribution is fairly spread across categories, with Others being the most common, followed by Comedy, Thriller, and Drama.
- The dayofweek distribution showed that most content is released on Friday and Wednesday, suggesting these are the preferred release days.
- Very few releases occur on Monday and Tuesday, indicating these days are rarely chosen for content launches.
- The season distribution is almost evenly balanced across Winter, Fall, Spring, and Summer, meaning the dataset does not have seasonal imbalance.

Standardizing Categorical Text Columns:

- The categorical variables genre, dayofweek, and season were standardized by removing extra spaces and converting text to lowercase.
- This ensures consistency in category labels and prevents duplicate categories caused by formatting differences.

Converting the binary variable into a meaningful label:

- The variable `major_sports_event` originally contained binary values 0 and 1.
- These values were converted into "No" and "Yes" labels to improve interpretability during exploratory data analysis and visualization.
- This transformation does not change the underlying information but makes the dataset easier to understand.

Checking for invalid numeric values:

- The numerical variables were checked using summary statistics.
- The average platform visitors is about 1.7 million, and ad impressions average around 1434 million.
- The average trailer views are about 66.9 million, with some trailers receiving significantly higher views.
- The average first-day content viewership is approximately 0.47 million.
- No invalid or non-positive values were found in the numerical variables.

Feature Engineering:

- The dataset already contains meaningful predictor variables such as visitors, ad impressions, trailer views, genre, season, and day of release.
- Since most content releases occur on Fridays and Wednesdays, creating additional timing features (such as weekend indicators) would not add significant value.
- Therefore, no additional feature engineering was performed, and the existing variables are used directly for modelling.

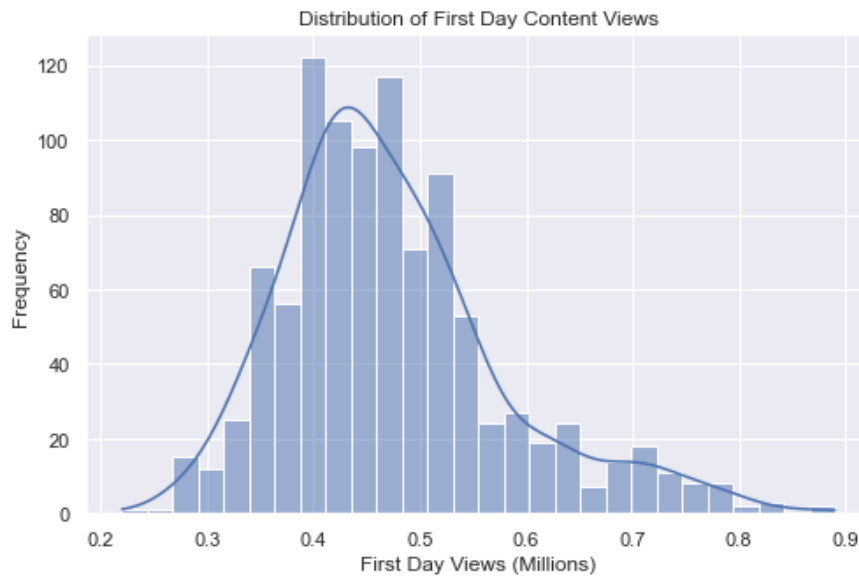
Exploratory Data Analysis:

Statistical Summary of the Dataset:

	count	mean	std	min	25%	50%	75%	max
visitors	1000.0	1.70429	0.231973	1.25	1.5500	1.70	1.830	2.34
ad_impressions	1000.0	1434.71229	289.534834	1010.87	1210.3300	1383.58	1623.670	2424.20
views_trailer	1000.0	66.91559	35.001080	30.08	50.9475	53.96	57.755	199.92
views_content	1000.0	0.47340	0.105914	0.22	0.4000	0.45	0.520	0.89

Univariate Analysis:

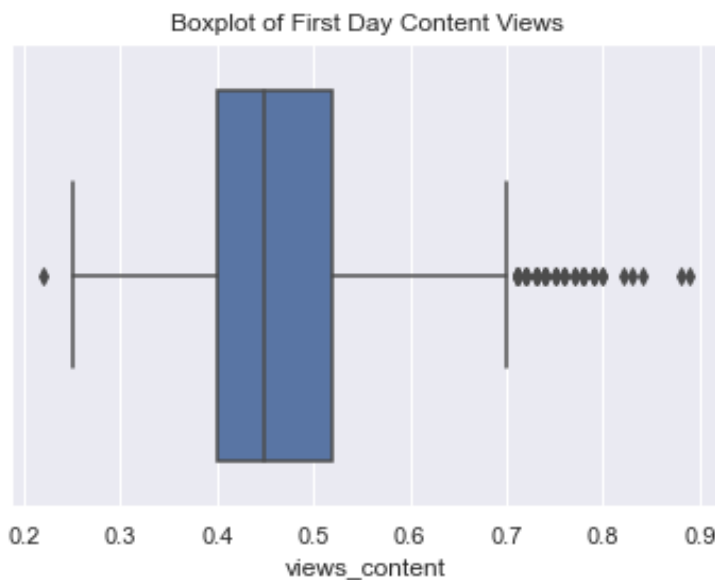
Distribution of First-Day Content Views (Target Variable):



Distribution of First-Day Content Views

- The distribution of first-day content views appears slightly right-skewed.
- Most content receives between 0.4 and 0.5 million views on the first day.

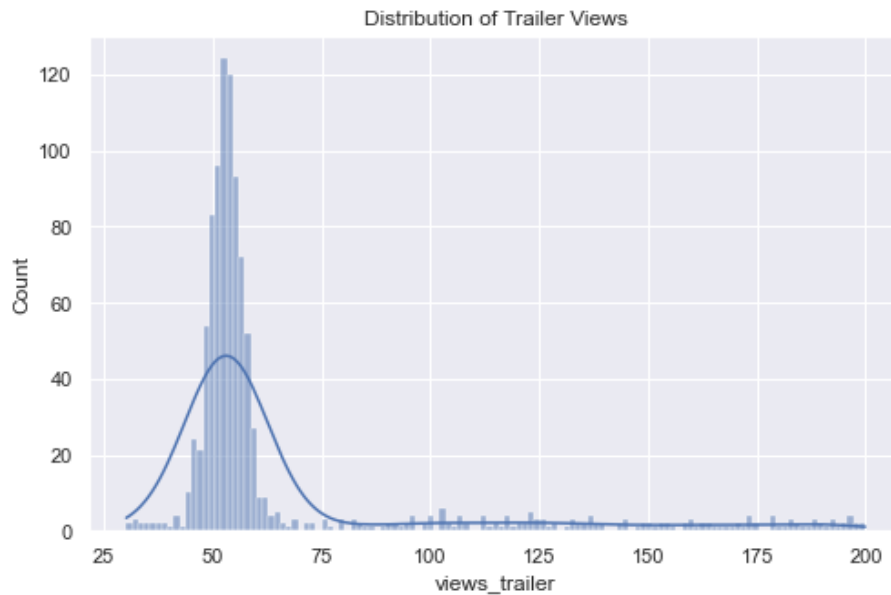
Boxplot of First-Day Content Views (Outlier Check):



Boxplot of First-Day Content Views

- The boxplot shows a few higher-value outliers, indicating that some content achieves significantly higher viewership than the majority.

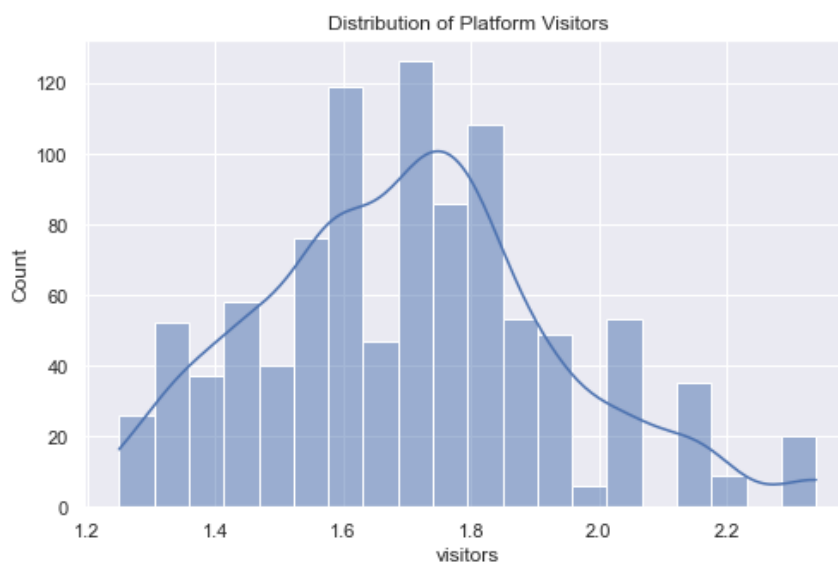
Distribution of Trailer Views:



Distribution of Trailer Views

- The distribution of trailer views is heavily right-skewed.
- Most trailers receive between 45 and 60 million views, while a small number achieve extremely high viewership.
- These higher values likely correspond to highly anticipated or heavily marketed content releases.

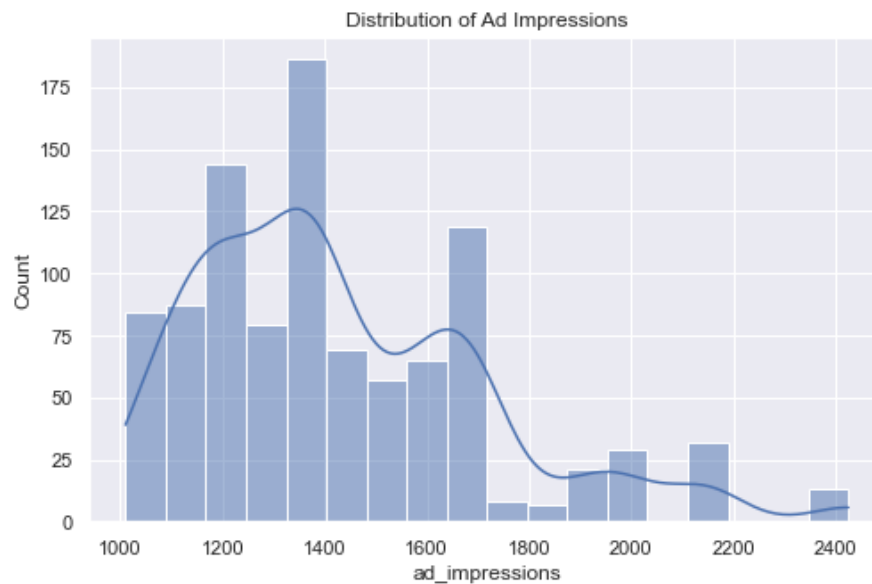
Distribution of Platform Visitors:



Distribution of Platform Visitors

- The distribution of platform visitors is approximately normal.
- Most weeks record between 1.5 and 1.9 million visitors, indicating relatively stable platform traffic.

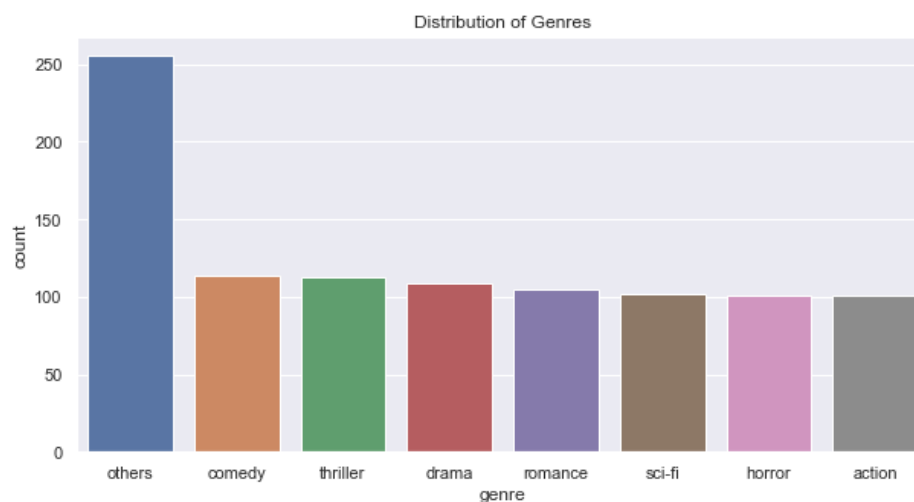
Distribution of Ad Impressions:



Distribution of Ad Impressions

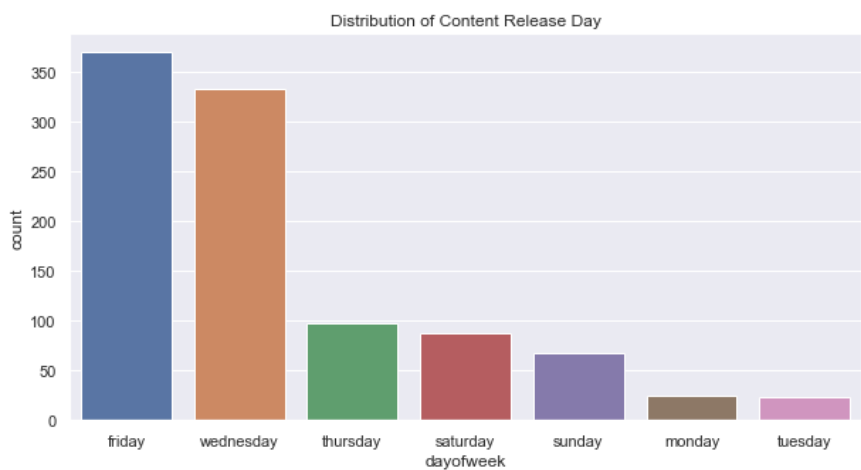
- The distribution of ad impressions is moderately right-skewed.
- Most content receives between 1200 and 1700 million ad impressions, while a few campaigns have significantly higher marketing exposure.

Distribution of Genres:



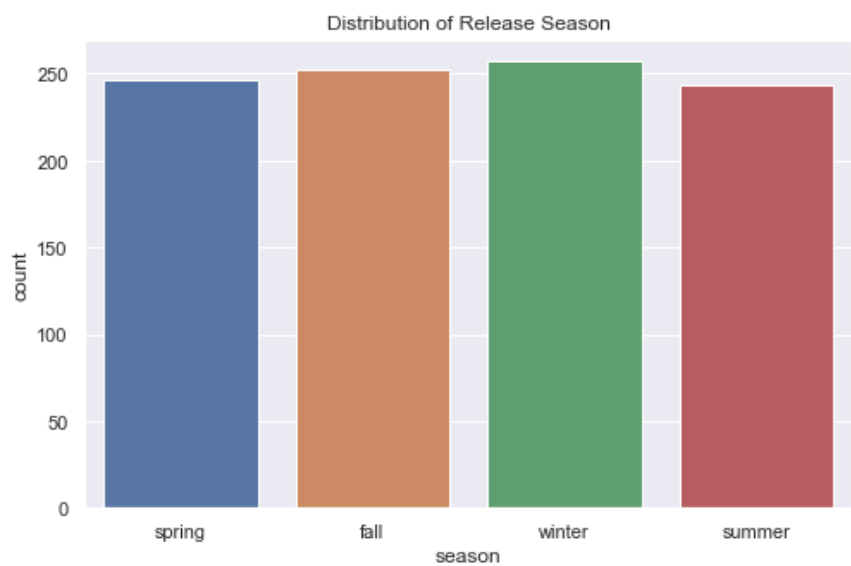
Distribution of Genres

Distribution of Release Day:



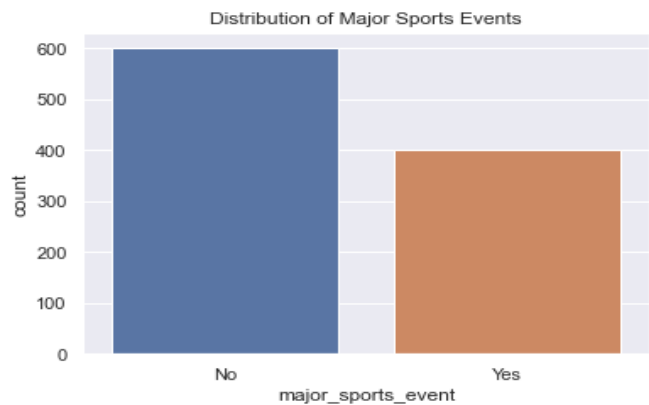
Distribution of Release Day

Distribution of Season:



Distribution of Season

Distribution of Sports Events:



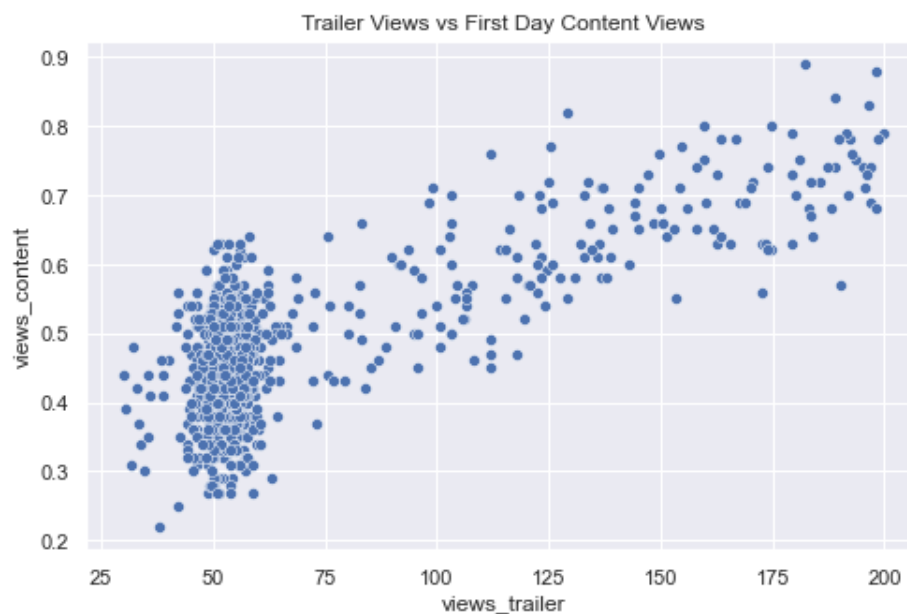
Distribution of Sports Events

Observations on Categorical Columns:

- The “Others” category dominates the dataset.
- Other genres such as Comedy, Thriller, Drama, Romance, Sci-Fi, Horror, and Action appear relatively balanced.
- This suggests the platform hosts a diverse mix of content types.
- The majority of content releases occur on Friday and Wednesday.
- Very few releases occur on Monday and Tuesday.
- This indicates that platforms strategically release content closer to the weekend period.
- Content releases are fairly evenly distributed across all four seasons.
- No strong seasonal bias is observed in the dataset.
- Most content releases (around 60%) occur without a major sports event.
- About 40% of releases coincide with major sports events, which could potentially impact viewership.

Bivariate Analysis:

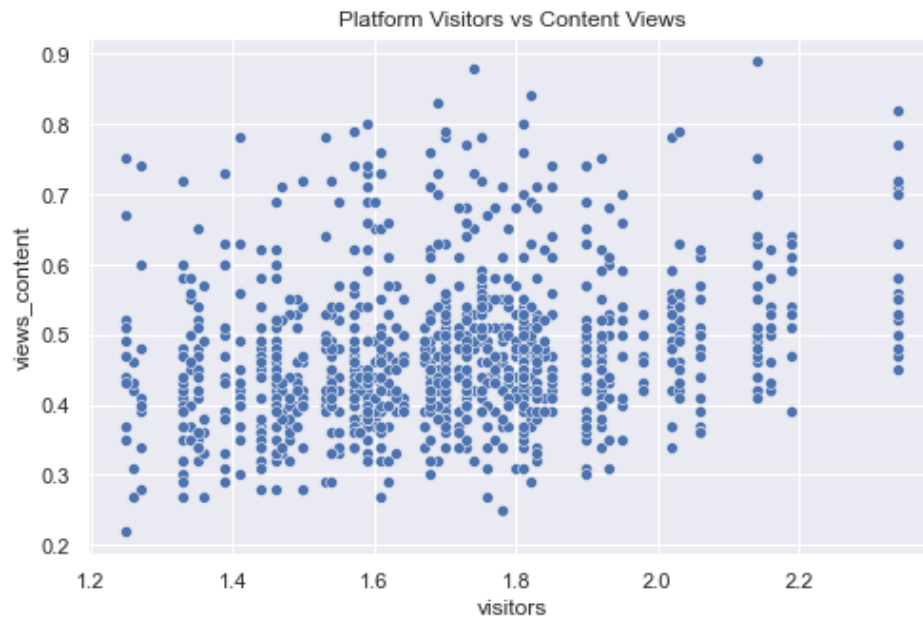
Trailer Views vs Content Views:



Trailer Views vs Content Views

- There is a strong positive relationship between trailer views and first-day content views.
- Content with higher trailer engagement generally receives higher first-day viewership.

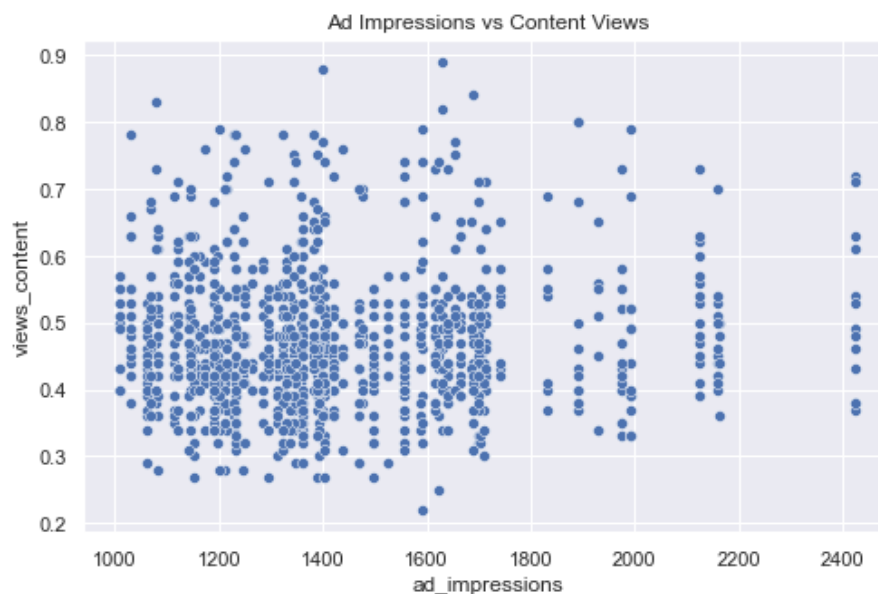
Platform Visitors vs Content Views:



Platform Visitors vs Content Views

- A weak positive relationship is observed between platform visitors and first-day content views.
- Higher platform traffic may contribute to increased content viewership, but the relationship is not very strong.

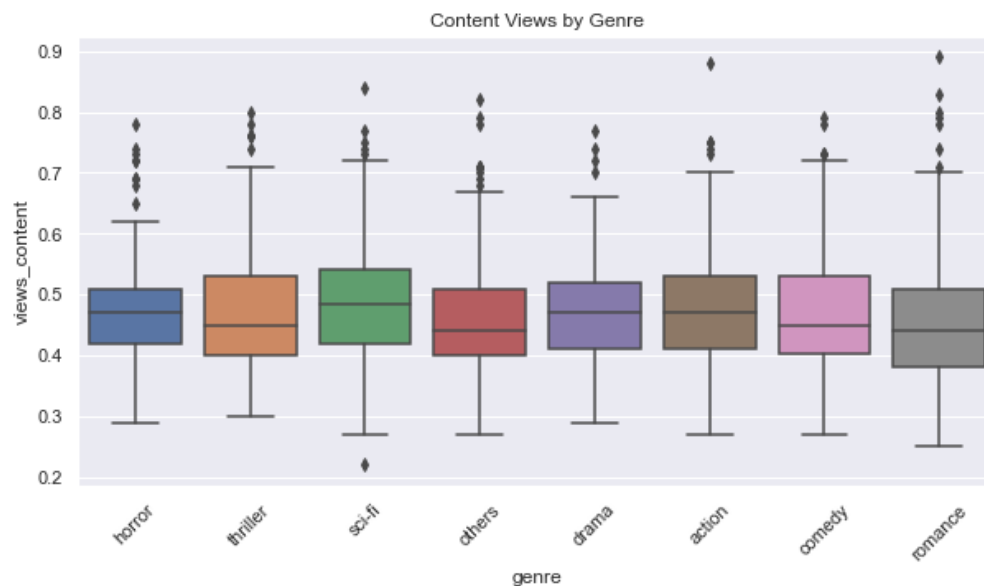
Ad Impressions vs Content Views:



Ad Impressions vs Content Views

- The scatter plot shows no clear linear relationship between ad impressions and first-day content views.
- This indicates that ad impressions alone may not strongly drive immediate viewership.

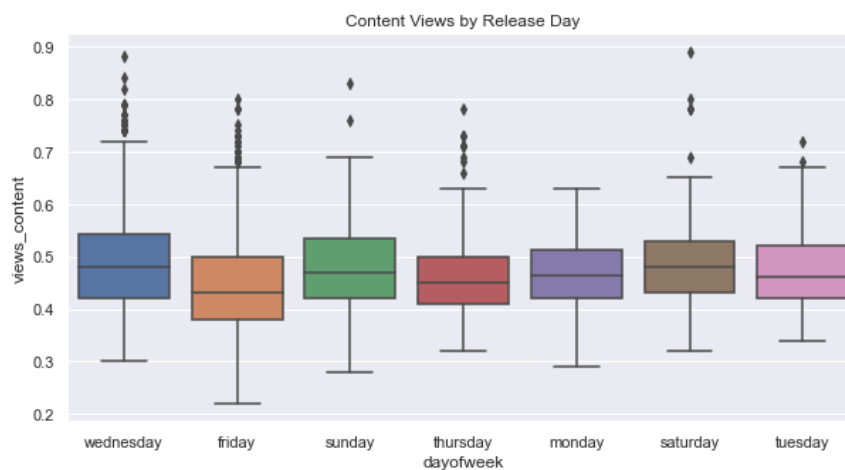
Content Views by Genre:



Content Views by Genre

- Viewership varies slightly across genres, but the median viewership is relatively similar across most categories.
- Some genres such as Sci-Fi, Action, and Drama show slightly higher upper ranges of viewership.
- Overall, genre does not appear to be a dominant factor influencing first-day views.

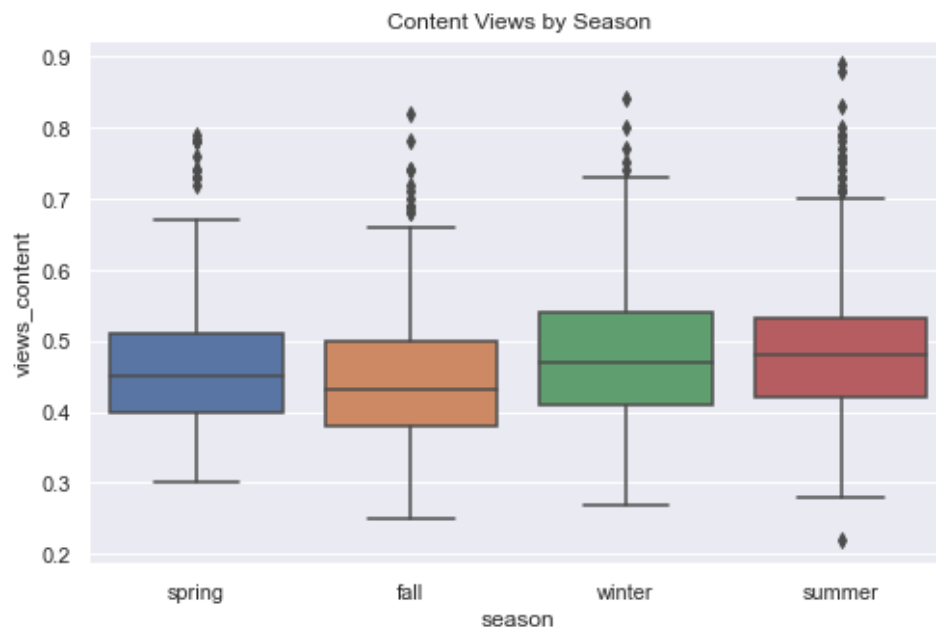
Content Views by Release Day:



Content Views by Release Day

- Viewership patterns differ slightly across release days.
- Wednesday, Saturday, and Sunday releases appear to show slightly higher median viewership compared to other days.
- This suggests that release timing may influence audience engagement.

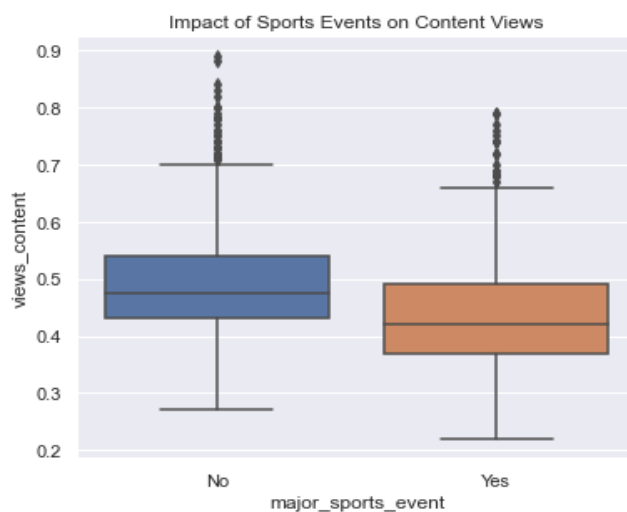
Content Views by Season:



Content Views by Season

- Seasonal variation in viewership is present but not extremely pronounced.
- Winter and Summer releases appear to show slightly higher median viewership compared to Spring and Fall.

Content Views vs Sports Events:



Content Views vs Sports Events

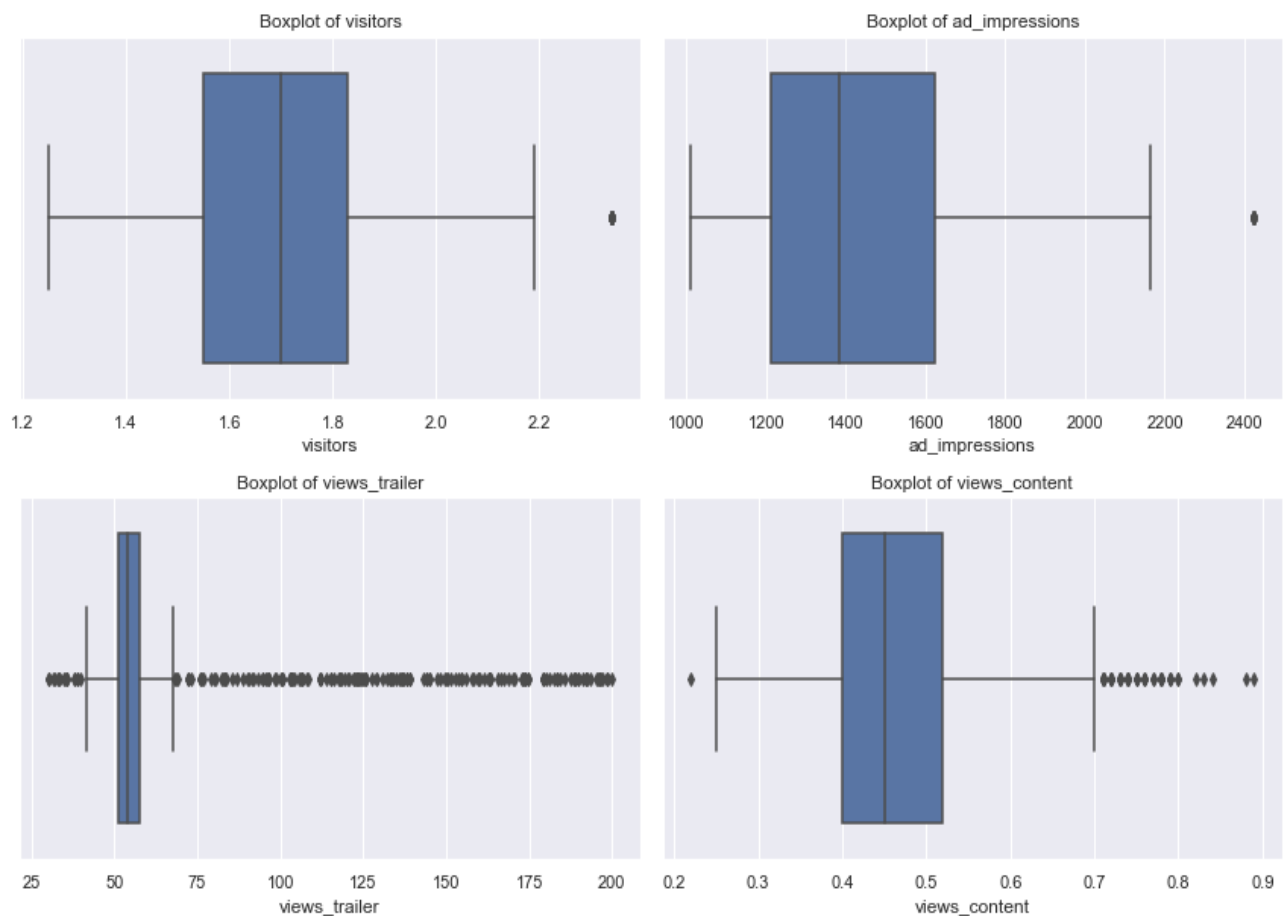
- Content released on days without major sports events tends to have slightly higher median viewership.
- When major sports events occur, viewership appears to decline slightly, possibly due to audience attention shifting to live sports.

Correlation Heatmap:



- Trailer views have the strongest positive correlation with content views (0.75).
- Platform visitors show a moderate positive correlation (0.26) with content views.
- Ad impressions show almost no correlation (~0.05) with first-day content views.
- Trailer popularity therefore appears to be the most important predictor of first-day viewership.

Boxplot for all numeric variables:



Visitors:

- The distribution of platform visitors appears relatively symmetric with most values concentrated between 1.5 and 1.8 million.
- There are very few outliers, indicating that platform traffic is generally stable.

Ad Impressions:

- The ad impressions variable shows a moderate spread with most values lying roughly between 1200 and 1600 million impressions.
- A few higher-value outliers are observed, indicating that some content received significantly higher marketing exposure.

Trailer Views:

- The distribution of trailer views is highly right-skewed with a large number of high-value outliers.
- Most trailers receive views around 50–60 million, while some trailers receive extremely high views (up to ~200 million).
- These extreme values likely correspond to highly promoted or very popular content trailers.

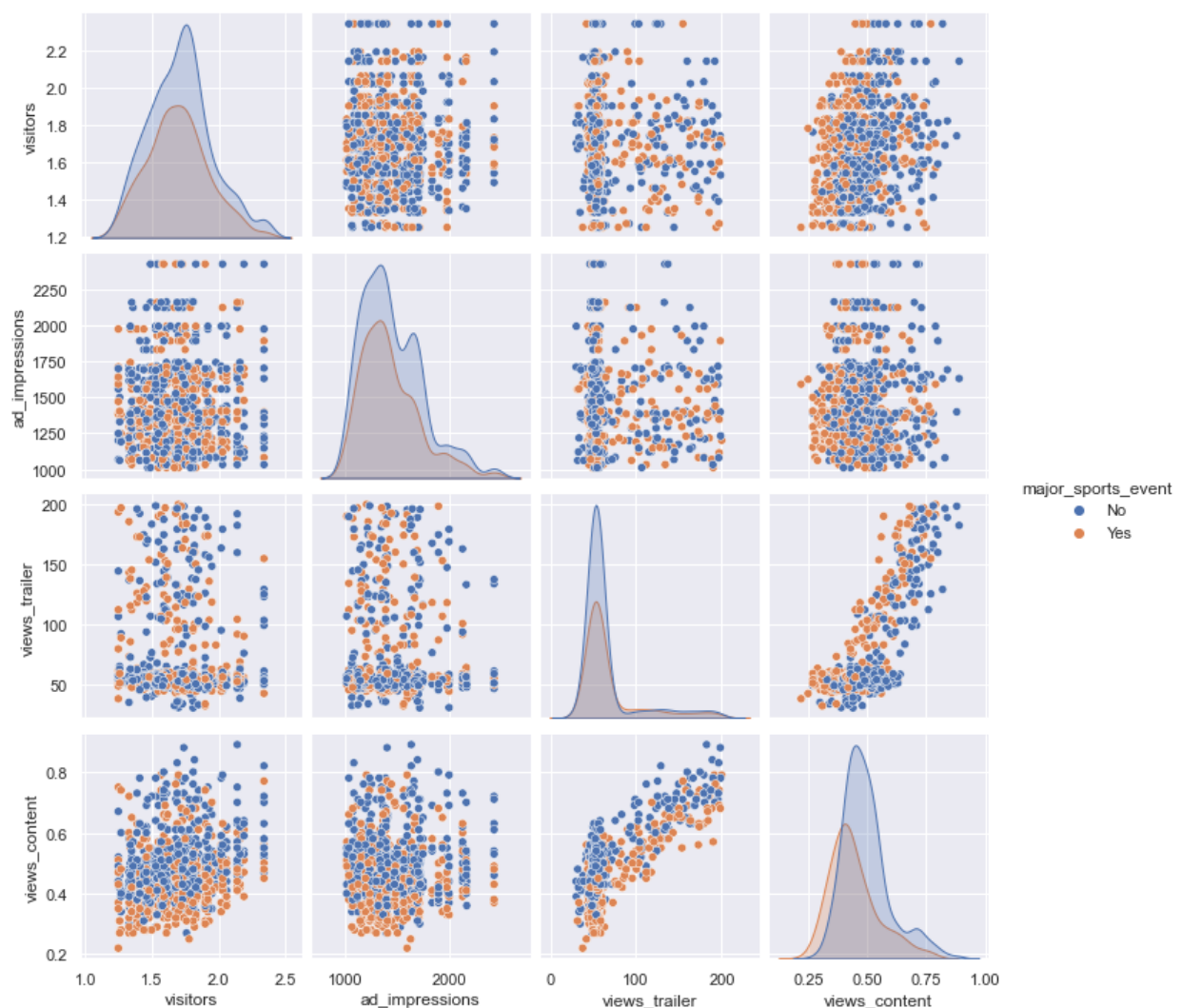
Content Views:

- First-day content views are moderately spread around 0.4–0.5 million views.
- Some higher-value outliers are present, indicating that certain titles significantly outperform the average content.
- These outliers are expected in entertainment platforms where a few releases become very popular compared to the majority.

Inference:

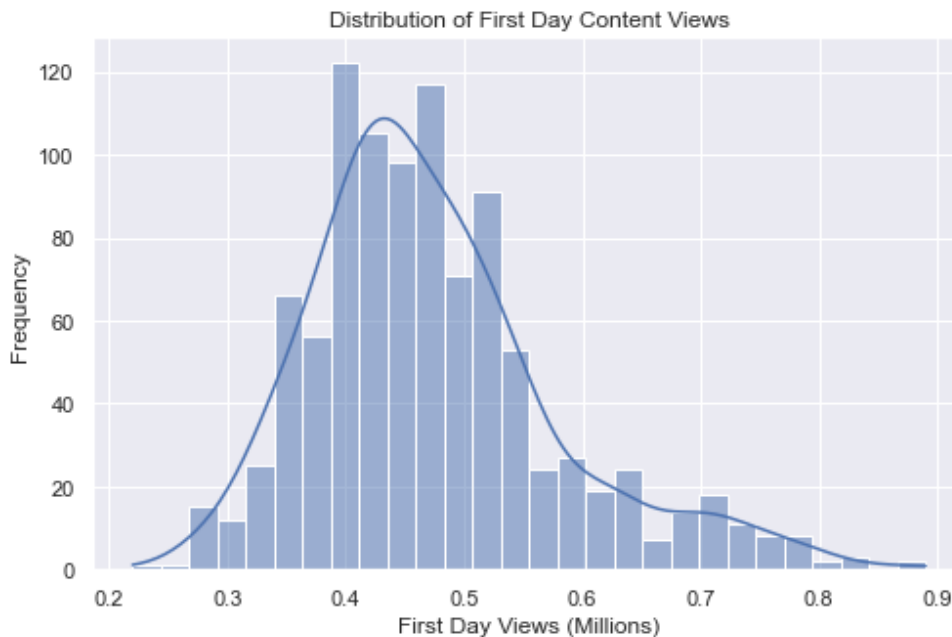
The boxplots show some extreme values, particularly in trailer views and first-day content views. However, these values likely represent highly popular or heavily marketed content rather than data entry errors as there are no negative values, impossible numbers or unrealistic orders of magnitude. Since the values fall within plausible ranges for digital media platforms, no outlier removal is performed at this stage.

Pairplot Analysis:



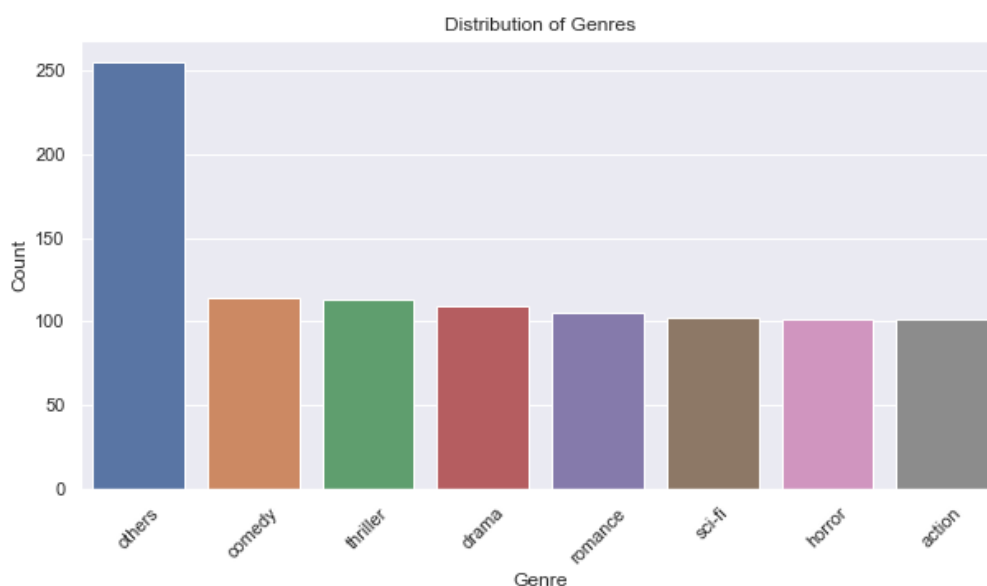
Answers to Key Questions

1) What does the distribution of content views look like?



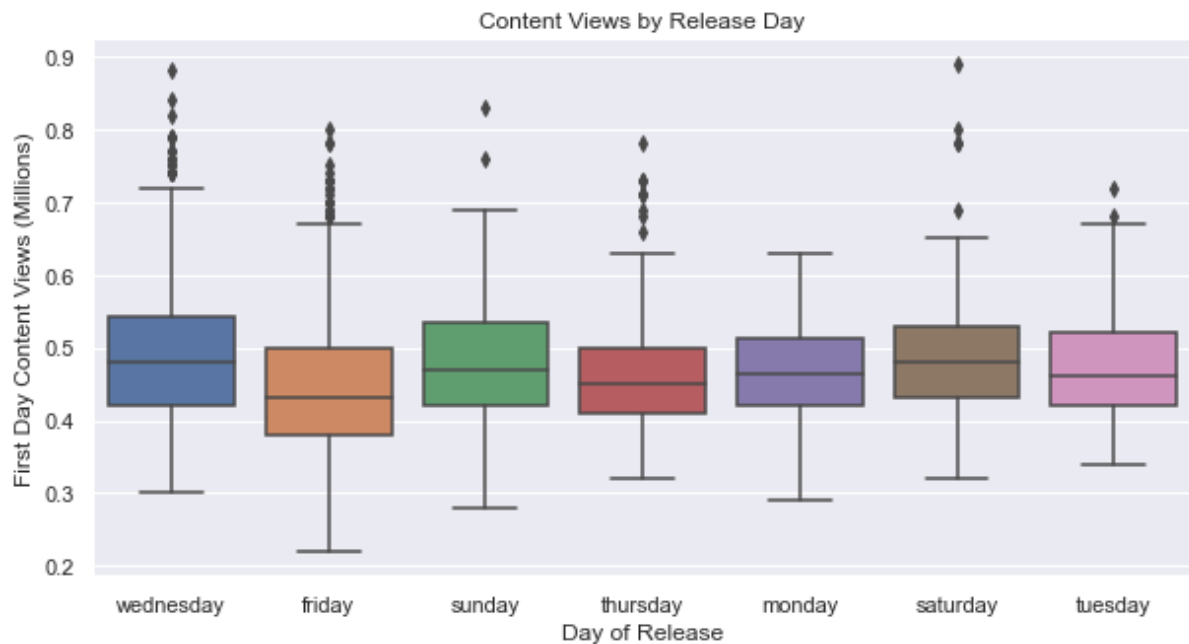
The distribution of first-day content views is slightly right-skewed, with most content receiving between 0.4 and 0.5 million views. A few titles achieve significantly higher viewership, which appear as outliers in the boxplot, indicating that some content performs exceptionally well compared to the majority.

2) What does the distribution of genres look like?



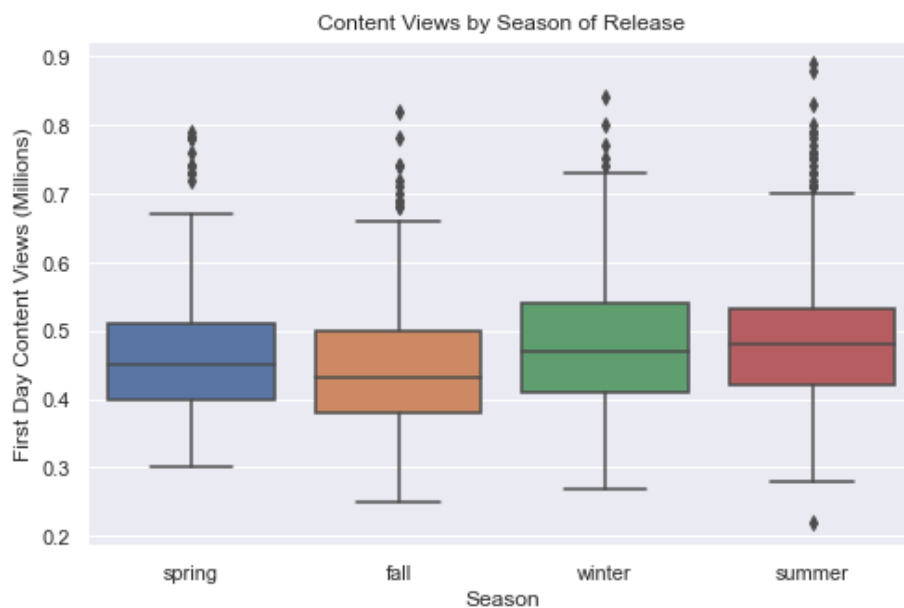
The dataset contains a variety of genres, with the "Others" category having the highest count of content releases. The remaining genres such as Comedy, Thriller, Drama, Romance, Sci-Fi, Horror, and Action are relatively balanced in distribution, suggesting that the platform offers a diverse mix of content types.

3) How does the viewership vary with the day of release?



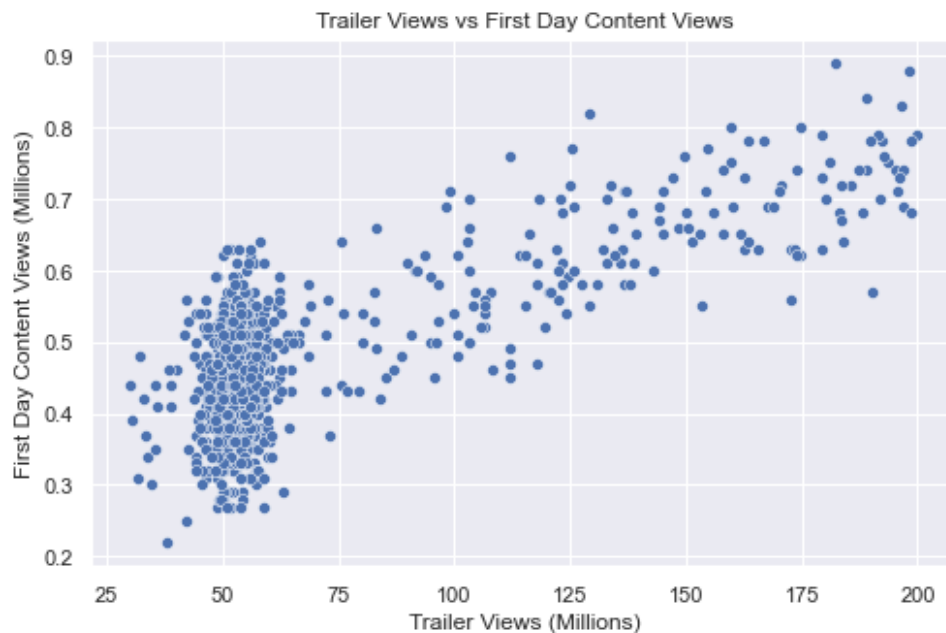
Viewership varies slightly depending on the release day. Wednesday, Saturday, and Sunday releases tend to show slightly higher median viewership, while Friday releases appear slightly lower in comparison. However, the differences are not extremely large, indicating that while release timing may influence viewership, it is not the dominant factor.

4) How does the viewership vary with the season of release?



Seasonal variation in viewership is present but not very pronounced. Winter and Summer releases show slightly higher median viewership, while Spring and Fall releases show marginally lower values. Overall, seasonality appears to have a moderate but not strong influence on first-day viewership.

5) What is the correlation between trailer views and content views?



	views_trailer	views_content
views_trailer	1.000000	0.753962
views_content	0.753962	1.000000

There is a strong positive correlation (≈ 0.75) between trailer views and first-day content views. The scatter plot clearly shows that content with higher trailer engagement tends to receive higher first-day viewership. This suggests that trailer popularity is one of the strongest drivers of content performance on the platform.

Model Building – Linear Regression:

- The objective of this analysis is to predict first-day content viewership (views_content) using the available predictor variables in the dataset.
- Since the dataset contains categorical variables such as genre, dayofweek, season, and major_sports_event, these need to be converted into dummy variables before fitting the regression model.
- To evaluate the model properly, the data will be split into train and test sets. A Linear Regression model will then be fitted on the training data, and its performance will be evaluated on both training and testing datasets.

Number of rows in train data = 700

Number of rows in test data = 300

OLS Regression Results			
=====			
Dep. Variable:	views_content	R-squared:	0.792
Model:	OLS	Adj. R-squared:	0.785
Method:	Least Squares	F-statistic:	129.0
Date:	Sun, 08 Mar 2026	Prob (F-statistic):	1.32e-215
Time:	19:29:30	Log-Likelihood:	1124.6
No. Observations:	700	AIC:	-2207.
Df Residuals:	679	BIC:	-2112.
Df Model:	20		
Covariance Type:	nonrobust		

Overall Model Fit:

- The model has an R-squared value of 0.792, which indicates that approximately 79.2% of the variation in first-day content viewership is explained by the predictor variables.
- The Adjusted R-squared value is 0.785, which is very close to the R-squared value. This suggests that the model is not overfitting and most variables contribute meaningfully to the model.
- The F-statistic is highly significant (p-value ~ 0), indicating that the overall regression model is statistically significant.

Model Coefficients with Column Names:

	Feature	Coefficient			
			11	genre_thriller	0.008708
0	const	0.060157	12	dayofweek_monday	0.033662
1	visitors	0.129451	13	dayofweek_saturday	0.057887
2	ad_impressions	0.000004	14	dayofweek_sunday	0.036321
3	views_trailer	0.002330	15	dayofweek_thursday	0.017289
4	major_sports_event_Yes	-0.060326	16	dayofweek_tuesday	0.022837
5	genre_comedy	0.009352	17	dayofweek_wednesday	0.047376
6	genre_drama	0.012625	18	season_spring	0.022602
7	genre_horror	0.009862	19	season_summer	0.044203
8	genre_others	0.006325	20	season_winter	0.027161
9	genre_romance	0.000551			
10	genre_sci-fi	0.013143			

Key Significant Predictors:

1] Visitors

- The coefficient for visitors (0.129) is positive and highly significant.
- This indicates that higher platform traffic in the past week strongly increases first-day content viewership.
- This variable appears to be one of the most important drivers of content performance.

2] Major Sports Events

- The coefficient for major_sports_event_Yes (-0.060) is negative and statistically significant.
- This indicates that major sports events reduce first-day content viewership, likely because viewers shift their attention to live sports events.

3] Effect of Release Day

- Several release days significantly impact viewership:
- Saturday (0.058) and Wednesday (0.047) show strong positive effects on viewership.
- Sunday (0.036) and Monday (0.033) also show positive impacts.
- This suggests that content released closer to weekends tends to attract more viewers.

4] Effect of Season

- Seasonal effects are also significant
- Summer (0.044) has the strongest positive impact on viewership.
- Winter (0.027) and Spring (0.023) also increase viewership compared to the reference season.
- This suggests that seasonal viewing behavior influences content consumption on the platform.

5] Genre Effects

- Most genre variables are not statistically significant, indicating that genre alone may not strongly determine first-day viewership.
- This suggests that platform traffic, marketing engagement, and release timing are more influential factors.

Inference:

- The condition number is relatively large, which may indicate possible multicollinearity among some predictors. However, further diagnostics such as Variance Inflation Factor (VIF) should be conducted to confirm this.
- The model suggests that platform traffic, trailer engagement, release timing, and external events (such as sports events) are the strongest drivers of first-day content viewership.

Testing Assumptions for Linear Regression Model:

Test For Multicollinearity:

	feature	VIF
0	const	99.679317
1	visitors	1.027837
2	ad_impressions	1.029390
3	views_trailer	1.023551
4	major_sports_event_Yes	1.065689
5	genre_comedy	1.917635
6	genre_drama	1.926699
7	genre_horror	1.904460
8	genre_others	2.573779
9	genre_romance	1.753525
10	genre_sci-fi	1.863473
11	genre_thriller	1.921001
12	dayofweek_monday	1.063551
13	dayofweek_saturday	1.155744
14	dayofweek_sunday	1.150409
15	dayofweek_thursday	1.169870
16	dayofweek_tuesday	1.062793
17	dayofweek_wednesday	1.315231
18	season_spring	1.541591
19	season_summer	1.568240
20	season_winter	1.570338

Observations:

- The VIF values for all predictor variables are well below 5, indicating that multicollinearity is not a concern in this model.
- The numerical variables visitors, ad_impressions, and views_trailer have VIF values very close to 1, which suggests they are highly independent of one another.
- The categorical dummy variables also have acceptable VIF values, mostly between 1 and 2.6.
- The high VIF for const can be ignored, as it is the intercept term and not interpreted like the predictor variables.

Inference:

The assumption of no multicollinearity is satisfied. Since there is no significant multicollinearity, we do not need to remove any variables at this stage.

Test Linearity and Independence of Errors:



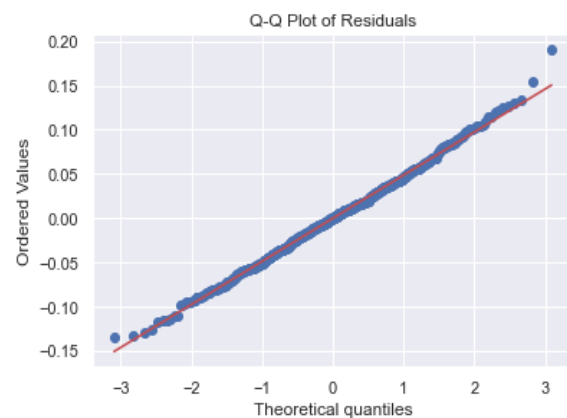
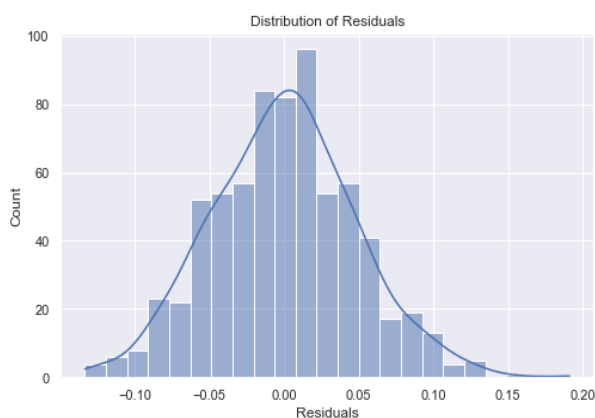
Observations:

- The Residuals vs Fitted Values plot shows that the residuals are randomly scattered around zero.
- There is no clear systematic pattern (such as a curve, funnel, or clustering) in the residual distribution.
- The spread of residuals appears fairly symmetric around the horizontal line at zero, which indicates that the model captures the underlying linear relationship reasonably well.
- Since the residuals do not show any obvious structure or dependency pattern, the independence of error terms assumption is also satisfied.

Inference:

- The assumptions of linearity between predictors and the target variable and independence of residuals appear to be satisfied for this regression model.

Test for Normality of Residuals:



Observations:

- The histogram of residuals shows a bell-shaped distribution, suggesting that the residuals are approximately normally distributed.
- The Q-Q plot indicates that most of the residual points lie very close to the reference straight line, which further supports the assumption of normality.
- The Shapiro–Wilk test gives a p-value = 0.284, which is greater than 0.05.

Inference:

- Since the p-value > 0.05, we fail to reject the null hypothesis that the residuals are normally distributed.
- Therefore, the normality assumption of linear regression is satisfied.

Test for Homoscedasticity (Constant Variance of Errors):

Observations:

- F-statistic = 1.144
- p-value = 0.1109
- Since the p-value > 0.05, we fail to reject the null hypothesis.

Inference:

- Because the p-value is greater than 0.05, there is no statistical evidence of heteroscedasticity.
- Therefore, the homoscedasticity assumption of linear regression is satisfied.

Model Performance Evaluation:

- Error Metrics (RMSE, MAE, MAPE), and goodness of fit metric (R-squared) were computed.

	RMSE	MAE	MAPE	R-squared
0	0.04853	0.038197	8.55644	0.791616

Training Performance Metrics

	RMSE	MAE	MAPE	R-squared
0	0.050603	0.040782	9.030464	0.766447

Test Performance Metrics

Observations:

- On the test dataset, the model achieved an RMSE of 0.0506, which is only slightly higher than the training RMSE.
- The MAE on the test data is 0.0408, indicating that the model maintains good predictive accuracy on unseen data.
- The MAPE is approximately 9.03%, which suggests that the model's predictions deviate from the actual values by about 9% on average.
- The R-squared value on the test set is 0.766, which is close to the training R-squared value.

Model Generalization:

- The performance metrics on the training and test datasets are very similar, indicating that the model generalizes well and is not significantly overfitting.
- The small difference between training and test performance suggests that the model is stable and reliable for predicting first-day content viewership.

Overall Model Performance

- The model demonstrates good predictive capability, explaining a large proportion of variance in the target variable.
- With a MAPE of around 9%, the model can predict first-day content views with reasonably high accuracy.

Actionable Insights and Business Recommendations:

Model Effectiveness:

The linear regression model explains a significant portion of the variation in first-day content viewership.

- Training R^2 : 0.79
- Test R^2 : 0.77
- The MAPE of approximately 9% on the test data indicates that the model can predict first-day content views with good accuracy. The training and test errors are close, suggesting that the model generalizes well and is not overfitting.

Significance of Predictors:

The regression results indicate that the following variables significantly influence first-day content views:

- Trailer Views: Trailer views have the strongest positive impact on content viewership. Higher trailer engagement leads to higher first-day views, highlighting the importance of effective trailer marketing.

- Platform Visitors: A higher number of visitors on the platform increases the likelihood of users watching newly released content, positively influencing viewership.
- Major Sports Events: The presence of a major sports event negatively affects content views, suggesting that audiences may shift their attention to live sports instead of streaming content.
- Day of Release: Content released on certain days, particularly Wednesdays and weekends, tends to receive higher first-day viewership.
- Season of Release: Seasonal effects are also observed, with summer and winter releases generally attracting higher engagement.

Key Takeaways for the Business:

- The analysis suggests that marketing exposure, platform traffic, and release timing are the key drivers of first-day content views.
- To improve viewership, the platform should:
 - Invest in strong trailer marketing campaigns
 - Increase platform traffic through promotions and recommendations
 - Schedule releases strategically on high-engagement days
 - Avoid launching major content during large sports events
 - Plan premium releases during high engagement seasons
- These strategies can help improve first-day content performance and overall audience engagement on the platform.